

Abhishek Kumar

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RESEARCH INTERESTS

Computational Neuroscience, Neuromorphic Computing, Spiking Neural Network, Neural Modeling, Machine Learning, Brain-Inspired Artificial Intelligence

EDUCATION

ABV-Indian Institute of Information Technology and Management Gwalior Gwalior, MP
MS (Artificial Intelligence and Data Science) *Aug. 2025 - Present*

University of Delhi Delhi
Bachelor of Science, Mathematics (H) *Nov. 2022 - Jun 2025*

EXPERIENCE/PROJECTS

Winter Research Internship [Offer letter](#) Dec. 2025 - Jan. 2026
Ion channel and Receptor biology lab, KSBS *Indian Institute of Technology, Delhi*

- Developed neuron simulations using **NEURON** to study ion-channel and spike dynamics.
- Implemented and modified biophysical neuron models using **hoc scripting** and explored ModelDB.
- Applied supervised learning models to molecular datasets for toxicity prediction, aligning with ML-driven drug discovery.

Aadiyogi Dec. 2022 - Jun. 2025
Team Captain *Deshbandhu College, UoD*

- Led the team in regional, state, and national-level yoga competitions and championships.
- Facilitated weekly practice sessions to discuss and master advanced Asanas, Pranayama & Meditation techniques.
- Acted as the primary liaison between team members, coaching staff, and competition officials.

PROJECTS

Ultra-Low-Power Edge Intelligence for Real-World Anomaly Detection Using SNN Ongoing
Spiking Neural Networks, Edge AI, Embedded Systems [Project info](#).

- Creating an event-driven anomaly detection framework to identify irregular patterns that indicate sewage blockages.
- Project currently under development; implementation details and results will be shared upon completion.

Multi-Objective Analysis of Spiking Neuron Models (Python) Jan. 2026
[GitHub](#) *Research Project*

- Built multi-objective evaluation framework for SNN models.
- Compared models on efficiency and biological realism.
- Simulated responses under varied stimulation protocols.
- Analyzed spike trains and membrane dynamics.

Spiking Neuron Simulations (Python, Brian2 / NEURON) Dec. 2025 - Present
[GitHub](#) *Self-Directed Practice Repository*

- Maintained a practice repository containing multiple implementations of spiking neuron models under varied conditions.
- Explored neuron behavior across different parameter regimes, input currents, and initial conditions.
- Used visualization of membrane potentials and spike trains to analyze model behavior.

Early-Stage Heart Attack Prediction (Python, ML)

[GitHub](#)

Oct. 2025

Machine Learning Project

- Built classification models for early heart attack prediction.
- Performed data preprocessing and feature engineering.
- Implemented CatBoost and XGBoost for improved accuracy.
- Optimized models using attention-based enhancements.

PUBLICATIONS

Abhishek Kumar, "Computational Neuroscience-Inspired Neuromorphic Computing: A Review of Models, Hardware Architectures, and Applications". (**Manuscript under review** / **Paper Presented**)

CONFERENCE AND SEMINARS

ICMSCI'26 [Certificate](#)

Feb. 2026

Paper Presented

PGDAV College, UoD

- * Presented a paper at International Conference on Mathematical Sciences and Computational Intelligence.
- * Topic: Computational Neuroscience-Inspired Neuromorphic Computing.
- * Analyzed TrueNorth, Loihi, SpiNNaker, and Neurogrid architectures.
- * Focused on energy-efficient AI and edge applications.

MATHSD'25 [Certificate](#)

Oct. 2025

Volunteering

ABV-IIITM Gwalior

- * Volunteered at MATHSD 2025, hosted by ABV-IIITM Gwalior.
- * Handled LaTeX typesetting and compilation of the official **Book of Abstracts**.
- * Assisted faculty and research teams in documentation and on-site coordination.
- * Gained insights on *Neuroscience-Inspired Scientific Machine Learning* through keynote sessions.

Samuchchayam 2025 [Certificate](#)

March 2025

Presented Paper

Deshbandhu college, UoD

- * Awarded first prize in the '**Datamatrix**' paper presentation competition.
- * Recognized for excellence in research, analysis, and public speaking at a competitive academic event
- * Successfully developed and presented a compelling paper on a data-centric topic to a panel of faculty and peers.

InitMath 2024 [Certificate](#)

March 2024

Mathematics Camp

Uttarkashi (U.K)

- * Selected for and completed the 'Initiation into Mathematics (InitMath) 2024' training program.
- * Participated in an intensive workshop sponsored by the prestigious **National Board for Higher Mathematics (NBHM)**.
- * Engaged in a 6-day program focused on foundational concepts in advanced mathematics, organized by **MTTS Trust**.

TRAINING AND CERTIFICATIONS

Computational Neuroscience – University of Washington [Certificate](#)

Nov. 2025 - Jan. 2026

- * Strengthened understanding of the intersection between Neural biology and computation.
- * Instructors: Rajesh P. N. Rao, Adrienne Fairhall

Introduction to Cognitive Psychology and Neuropsychology - University of Cambridge Jan. 2026 - Present

- * Learned theories, models and findings of cognitive functions, including memory, attention and perception.
- * Instructors: Dr. Giulia Mangiaracina

Linear Algebra for Machine Learning – LDRP Institute of Technology & Research [Certificate](#) Dec. 2025

- * National Short-Term Training Program
- * Hands-on experience on PCA, SVD, and image processing.

ACHIEVEMENTS & AWARDS

National Yoga Championship Runner up [Certificate](#) Sept. 2023

- * Championship held in Haridwar and organised by **World Fitness Federation of Yogasana Sports, India**

Certificate of Commendation by Chief Minister of Uttarakhand [Certificate](#) June 2020

- * Award presented to students with outstanding achievements in academics.

SKILLS & INTERESTS

Languages: Python, MySQL, R

Tools: Power BI, Excel, Mathematica, MATLAB

Frameworks & Libraries: Numpy, Pandas, Matplotlib, Seaborn, Scikitlearn, PyTorch, snnTorch, NEURON

Domains: Computational Neuroscience, Machine Learning, Deep Learning, Neural Network

Others: Kaggle, Apache Kafka , Git, GitHub